

1. Purpose

This procedure describes the removal and installation of the hydraulic pump assembly installed in the HX-20 helicopter hydraulic system. The procedure applies to standard maintenance configuration aircraft and includes additional notes for MK2 bracket installations.

2. References

Reference documents include HX20-AMM-00-00-00 General Safety Practices, HX20-AMM-20-10-00 Standard Torque Values and HX20-IPD-29-10-00 Illustrated Parts Data.

3. Safety precautions

Disconnect aircraft electrical power before beginning maintenance work. Release hydraulic system pressure before disconnecting hydraulic lines. Wear approved protective gloves and eye protection during maintenance procedures. Ensure all removed components are tagged and stored according to maintenance procedures.

4. Tools and materials

Required tools include a calibrated torque wrench, inspection flashlight, hydraulic pressure release tool, cleaning cloths, inspection mirror and standard maintenance toolkit. Approved hydraulic fluid HX-11 and replacement seals must be available before installation.

5. Access procedure

Open access panel 32-A and inspect surrounding hinges and fasteners for signs of damage or corrosion. Remove contamination from surrounding surfaces before disconnecting hydraulic lines. Verify that all work area lighting is adequate before continuing.

6. Hydraulic pump removal

Disconnect electrical connector P-01 from the hydraulic pump assembly. Disconnect hydraulic return and pressure lines and install protective caps immediately after removal. Remove four mounting bolts securing the hydraulic pump assembly to the support bracket. Carefully remove the hydraulic pump from the bracket and transfer the assembly to a clean inspection surface. Inspect surrounding hydraulic fittings for signs of leakage or contamination.

7. Inspection

Inspect removed seals for wear, cracking or deformation. Inspect hydraulic fittings and electrical connector insulation for signs of heat damage or contamination. Replace damaged seals before installation. Verify that mounting hardware meets current approved maintenance specifications.

8. Hydraulic pump installation

Position hydraulic pump assembly onto support bracket and install mounting bolts by hand before tightening to approved torque values. Reconnect hydraulic pressure and return lines. Reconnect electrical connector P-01. For MK2 helicopters use bracket type B and verify alignment before tightening mounting hardware. Remove all protective caps before operational testing.

9. Operational test

Reconnect aircraft electrical power and pressurize the hydraulic system. Inspect hydraulic lines and fittings for leakage during operation. Verify stable hydraulic pressure during flight control surface checks. Record maintenance actions in aircraft maintenance log.

10. Field maintenance observations

Previous maintenance revisions contained inconsistent terminology for the hydraulic pump assembly and repeated safety information across multiple sections. Several field maintenance reports identified incorrect torque values following overnight maintenance work. Additional review of maintenance documentation was recommended to improve navigation and procedural consistency.